# HARSHA DURAIRAJ

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## PROFESSIONAL SUMMARY

Engineering student specializing in machine learning and robotics, with hands-on experience in EEG/EMG signal processing, biomedical AI, and autonomous navigation through academic research, industry internship, and applied projects. Skilled at building end-to-end ML pipelines, deploying models on real hardware, and translating research into working systems. Strong problem-solving mindset with the ability to quickly learn new technologies and deliver reliable solutions in fast-paced engineering environments.

## EDUCATIONAL QUALIFICATION

**B.Tech Robotics & Artificial Intelligence (Pursuing)** *June 2027*
*SASTRA Deemed University | CGPA: 8.24*

**Higher Secondary Certificate** *June 2023*
*Muvendar Matric Higher Secondary School | 89.66%*

## TECHNICAL SKILLS

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|---|---|
| **Programming Languages:** | Python, C, MATLAB |
| **Frameworks & Libraries:** | PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV |
| **Tools:** | ROS2, Arduino IDE, Google Colab, VS Code |
| **AI / ML Techniques:** | Machine Learning, Deep Learning, Feature Extraction, Model Optimization, Explainable AI (SHAP), EEG/EMG Signal Processing |
| **Web Technologies:** | HTML |

## SOFT SKILLS

Leadership • Problem Solving • Teamwork • Time Management • Creativity & Innovation • Communication

## INTERNSHIP / EXPERIENCE

**QNeuro India Pvt. Ltd** *Dec 2025*
*Machine Learning Intern*

- Designed and implemented end-to-end machine learning pipelines involving data preprocessing, feature engineering, model development, and performance evaluation.
- Optimized feature extraction and model tuning techniques, achieving 97% classification accuracy while improving model robustness and generalization.
- Contributed to AI-driven biomedical projects by applying machine learning techniques to solve real-world healthcare challenges in a collaborative development environment.

## PROJECTS

**CADAI – EEG-Based Cognitive Workload Classification System** *May 2026*

- Developed a hybrid CNN-GCN model using 32-channel EEG signals, achieving 88.89% LOSO-CV accuracy for real-time cognitive workload classification.
- Applied WGAN-GP data augmentation and Explainable AI (SHAP) with RAG integration for improved generalization and interpretability.
- Technologies: Python, PyTorch, EEG signal processing, SHAP, WGAN-GP, Graph Neural Networks.

**Trajectory Tracking and Obstacle Avoidance using TurtleBot3 Burger** *May 2026*

- Built a closed-loop controller enabling TurtleBot3 to accurately follow straight, circular, and pentagon paths using odometry feedback.
- Designed a hybrid APF–Bug2 navigation strategy for real-time obstacle avoidance and path recovery in dynamic environments.
- Deployed and validated the system on real hardware after simulation testing in ROS2/Gazebo, quantifying tracking precision using IAE, ISE, and MSE error metrics across all three path types.
- Technologies: ROS2, Gazebo, RViz, Python, LiDAR.

**Adaptive Exoskeleton with EEG-EMG Fatigue Detection** *Nov 2025*

- Developed ML model integrating EEG and EMG signals for real-time fatigue detection.
- Applied advanced signal processing and feature extraction techniques to improve detection accuracy and model robustness across varying fatigue conditions.
- Technologies: Python, EMG/EEG signal processing, Scikit-learn, feature engineering.

## RESEARCH WORK

**CTGAN-Augmented Explainable AI Framework for Endometriosis Risk Stratification** *Ongoing*

- Developing an explainable machine learning framework for early non-invasive endometriosis risk stratification using CTGAN-based data augmentation and RAG-LLM-driven clinical decision support.

**EEG-Driven Physical Load Detection During Human Walking** *Published*

- Developed machine learning models for EEG-based physical load detection during human walking.

## CERTIFICATIONS

**Certificate of Presentation – RIACT 2026 | VIT Chennai** *Feb 2026*

- "EEG-Driven Physical Load Detection During Human Walking for Robotic and Automation Systems" presented at RIACT 2026, organized by VIT Chennai.

**Generative AI: Prompt Engineering Basics | IBM** *Dec 2025*

- Gained practical understanding of prompt design strategies for large language models.

**Cloud Computing | NPTEL (IIT Kharagpur)** *Nov 2025*

- Gained practical knowledge in virtualization, cloud service models (IaaS, PaaS, SaaS), resource provisioning, and scalability.

## LANGUAGES KNOWN

Tamil (Native Proficiency), English (Professional Working Proficiency)